AJ MENDES

INDUSTRIAL SOFTWARE | DATA SYSTEMS | SECURE ENGINEERING

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PROFILE

Colorado School of Mines computer science student combining software, cybersecurity data research, and hands-on industrial systems experience. Builds testable Python and Java tools, works comfortably from technical documentation, and communicates results across engineering teams.

EDUCATION

Colorado School of Mines

B.S. Computer Science | Minors in Military Science and Quantum Engineering | Cyber Defense Education Certificate | Expected May 2027 | GPA: 3.5

EXPERIENCE

Undergraduate Data Science Researcher

The Data Mine, Purdue University | Jan 2026 - May 2026

- Built Python evaluation tools for K-means clustering on labeled synthetic cybersecurity data, reporting accuracy, adjusted Rand index, and normalized mutual information across controlled experiments.
- Tested robustness across noise, cluster-count, and feature-variation sweeps; observed approximately 98% cluster-assignment accuracy in labeled tests with up to 20 clusters.
- Developed centroid-matching and drift-visualization workflows using PCA, UMAP, and t-SNE; documented and presented findings with Purdue, Data Mine of the Rockies, and CrowdStrike collaborators.

Electrical Control Panel Technician

Xpect Solutions, Longmont, CO | Jun 2025 - Aug 2025

- Assembled and wired industrial electrical control panels from schematics, wiring diagrams, and technical drawings.
- Routed, bundled, terminated, and labeled conductors to NEC and company standards while performing detail-focused quality checks across concurrent builds.

Construction and General Labor

Formit Right Concrete, Longmont, CO | Summers 2020 - 2024

- Completed five summers of full-time concrete, site-preparation, material-handling, and landscaping work in changing jobsite conditions and early-shift schedules.

SELECTED TECHNICAL WORK

Secure Chat Application

Python | TCP sockets | RSA-OAEP | AES-GCM

- Built a Python TCP client-server chat application using RSA-2048/OAEP key exchange and AES-256-GCM authenticated encryption; the relay forwards ciphertext without access to plaintext.

MarketPulse AI

Python | scikit-learn | TensorFlow/Keras | Streamlit

- Built a Python ML research pipeline that engineers returns, volatility, moving-average, volume, RSI, and MACD features for seven equities against SPY.

SKILLS AND LEADERSHIP

Languages: Python, Java, C++, SQL, R, TypeScript/JavaScript, HTML/CSS

Tools: Git, Linux, pandas, NumPy, scikit-learn, JUnit 5, React, Streamlit

Additional: Army ROTC, Mines ORESEC/CCDC, Secret security clearance